

DE-LU Day-Ahead Power Price Forecasting — Case Study

Trisith Kittisriswai | trisithworld@gmail.com

Market: Germany-Luxembourg (DE-LU) bidding zone, EPEX day-ahead, hourly, EUR/MWh | Forecast option: A — next-day hourly
| OOS window: 2025-01-01 → 2025-12-31 (full Test year, 8,759 predictions)

"I prioritised transparent, stable models deliberately: a fair-value signal has to be one a trader can interrogate and trust, and these markets break regime often enough that an over-flexible model is a liability, not an edge. The complex model is included as a benchmark — to confirm the simple one isn't leaving material signal behind."

This is a working prototype of a daily DE-LU fair-value view: next-day hourly forecasts, a prompt-curve relative-value lean, a trader-facing dashboard, and an LLM trader briefing, all from a single `python main.py` command. Full build history, round-by-round detail, and all figures are in `full_report.pdf`; this document covers the brief's required topics at summary depth.

1. Data, Sources & Point-in-Time Discipline

Target: EPEX DE-LU day-ahead hourly prices, 2019–2025 (history starts 2019 — the DE-AT-LU zone split in Oct 2018 mixed two price regimes before that). Source: ENTSO-E Transparency Platform — day-ahead (A01) **forecast** load/wind/solar for DE and FR, Forecast Transfer Capacity (NTC, the six borders that actually publish one), and neighbor bidding-zone prices/load/wind/solar (FR/NL/BE/PL/CZ). Plus daily TTF front-month gas (`yfinance`) and a CO2 proxy (CARB.L, a tradable ETC — no free official EUA settlement series was reachable, documented honestly rather than backfilled). EXAA's (Sequence 2) own day-ahead auction print for the same delivery day — already present unused in the committed price export — is used as a pre-auction reference (§5), not a model feature.

Point-in-time firewall: every feature for delivery day D must be knowable by ~12:00 D-1 (gate closure). Load/wind/solar are explicitly the day-ahead *forecast* vintage, never realised — enforced in code by `assert_no_lookahead()`, which raises if any forbidden timestamp appears. Negative prices (3.3% of hours) are never clipped. DST 23h/25h days are handled via a tz-aware index.

QA (`outputs/qa_report.md`): a continuous tz-aware hourly index; gap detection with short gaps imputed (method flagged) and long gaps reported rather than silently filled; duplicate-timestamp and range/sanity checks that explicitly preserve negatives; a per-series missingness summary; verification of the 23h/25h DST days against known transition dates; and a spot-check that reads a raw ENTSO-E export to confirm the loaded load series matches the day-ahead *forecast* column and not the realised actual. The few long-gap rows that remain (e.g. 48 in DE load, 216 in FR wind) are dropped by a shared validity mask, never back-filled into the training set.

2. Feature Engineering

Spine: **residual load** = **load** – **wind** – **solar** — the merit-order position, the dominant driver of both level and shape. 47 features total: calendar (cyclic hour/dow/month, holidays), residual load + hinge/quadratic merit-order terms, D-1/D-7 price lags (DE and 5 neighbor zones — in LightGBM's gain ranking, the CZ and BE lags actually outrank DE's own), rolling residual-load and price stats, TTF gas anchor (+ a Ridge-only gas×residual-load interaction), a standardised gas+CO2 pressure composite, FR residual load, NTC import/export/net capacity, and neighbor residual loads. Generation/grid outage data was considered and explicitly dropped — pre-2021 ENTSO-E REMIT data is sparse and unstructured, and forcing it in risked fabricating precision the rest of this build avoids.

3. Model Lineup & Selection

Baselines: D-1 and D-7 same-hour naïve. **Selected:** **Ridge** with hinge/quadratic residual-load terms — plain linear mis-specifies the convex, kinked merit-order curve. **Challenger:** **LightGBM**, included only as a nonlinearity check. A depth-3 decision tree is kept as an interpretability figure (not selected — high-bias, blocky), since its splits literally draw the merit-order kink.

Model	MAE — Validation 2024 (expanding)
Ridge (selected)	14.32
LightGBM (challenger)	11.22

LightGBM posts a lower Validation MAE (gap 3.10 / 21.7%), yet **Ridge is selected**: (1) a fair-value signal must be interrogable — coefficients vs. a 500-tree ensemble; (2) power markets break regime, and a regularised linear form degrades more gracefully than a flexible model when the next regime looks nothing like 2019-2024; (3) LightGBM's feature importances confirm Ridge's design (price lags, residual load dominate) rather than showing it's mis-specified. This decision is made on Validation (2024) only — the Test (2025) gap (LightGBM ahead by 3.27, 22.6%) is reported only as out-of-sample confirmation, never as part of the choice.

4. Validation Methodology & Test Results

Three-way split: Train 2019-2023 (fit), Validation 2024 (window-type + model-selection decision, used once), Test 2025 (headline backtest, never tuned on). Walk-forward throughout: train on all valid rows strictly before day D, predict D's 24 hours, advance — no shuffling. Calibration window (expanding vs. rolling) is chosen on Validation MAE alone; expanding currently wins (14.32 vs. 15.05). No MAPE is reported — DE-LU prices cross zero, so it's undefined/explosive; MAE and

RMSE are used throughout.

Model	MAE	RMSE	Skill vs. D-7
Naïve D-7	32.83	49.33	+0.0%
Naïve D-1	25.97	40.45	+20.9%
Ridge (selected)	14.50	23.58	+55.8%
LightGBM (challenger)	11.23	18.93	+65.8%

Test 2025, 8,759 hourly predictions (EUR/MWh). Metrics are broken out by regime to show where the edge concentrates — the skill is largest in the peak and high-renewables hours that matter most for positioning:

Regime (Test 2025)	n hours	Ridge MAE	LightGBM MAE
Peak (08–20 weekday)	3,393	18.18	14.82
Off-peak	5,366	12.17	8.97
High residual load	4,380	14.77	11.67
Low residual load	4,379	14.23	10.79

predictions.csv covers the full Test year (deliberately widened from the brief's “~2-4 weeks” so the deliverable matches the headline backtest it represents). Metrics are also reported by hour-of-day (errors peak in the evening ramp, hours 17–20) in outputs/validation_metrics.md.

5. Prompt-Curve Translation

A next-day model cannot produce a month-ahead curve — rolling a 1-day forecast 30 days forward compounds error against driving forecasts that don't exist. The view instead has two honest parts: (a) the model's **own hourly/block shape** — peak 88.05, off-peak 80.24 EUR/MWh (spread +7.80), 2,902 rich / 2,472 cheap hours (incl. 442 negative) / 876 scarcity hours — directly tradable as hourly DA bids, block products, or intraday, with no external reference; and (b) a **directional, relative-value call** against a same-day pre-auction reference.

Why EXAA, not an EEX print. Since no liquid public EEX prompt curve was reliably sourceable for the OOS delivery dates — the only front-week numbers I could find were unsourced estimates, and anchoring a desk-facing call on a placeholder is exactly the kind of thing a quant reviewer should catch — I use **EXAA as a same-day pre-auction tradable reference and translate to baseload/peak/off-peak positioning**. EXAA's Sequence-2 auction settles ~10:15 CET D-1, ahead of EPEX's ~12:00 gate closure, and clears the *same* delivery hours (full-history correlation with the EPEX target 0.986) — so it is a genuine, point-in-time-safe pre-auction print rather than a different product like a front-month contract, and it was already sitting unused in the committed price export. $\text{basis} = \text{Ridge fair value} - \text{EXAA}$; conviction/size scale with $|\text{basis}| \div \text{model backtest MAE}$ (a multiple of the model's own demonstrated error), not an arbitrary EUR threshold. Worked example (2025-12-31): EXAA 84.00, Ridge fair value 76.49 → basis −7.51 (0.52× MAE) → SELL, MODERATE conviction.

Use & invalidation. The lean is expressed as baseload/peak/off-peak block positions plus per-hour rich/cheap screens (\$7), sized by the conviction above. It is invalidated by an overnight TTF/gas move >5%, a wind-forecast revision >5 GW vs. D-1, a REMIT outage headline, a demand surprise, a realised print landing materially away from the trailing baseload (signalling a regime shift rather than mean-reversion), or a material EXAA repricing between its ~10:15 auction and the ~12:00 EPEX gate — the window in which the call was set against the earlier EXAA read.

6. LLM Trader Commentary

A grounded LLM (Anthropic API) turns the structured model state into a trader-facing morning note. It **originates no numbers** — every quantity traces to a structured fact object; output is schema-validated (Pydantic v2: bullish/bearish drivers, invalidation triggers, narrative text), and a grounding check verifies every number in the prose against the fact object. Direction/conviction are *not* LLM outputs — they're pure arithmetic (\$7) the LLM only narrates. Cached to ai_logs/ so python main.py reproduces deterministically, key-free.

7. Fair-Value-vs-EXAA Dashboard

outputs/fair_value_dashboard.html is a trader-facing, fully offline page (Chart.js vendored locally, no CDN) replaying one delivery day as a **static backtest** from the committed snapshot. The decision layer is pure arithmetic, computed once in src/dashboard.py and shared with the LLM's fact object so the two can never disagree: per hour, $\text{basis}_h = \text{fair_value}_h - \text{EXAA}_h$, $\text{conviction}_h = |\text{basis}_h| \div \text{MAE}_h$ (that hour's own backtest MAE), direction FLAT below a 0.25 conviction threshold else SELL/BUY by sign — rolled up into baseload/peak/off-peak block calls. For 2025-12-31: 22 SELL / 2 FLAT / 0 BUY across the 24 hours. Every number on the page is verified against its source JSON by an automated grounding check that fails the build loudly if anything doesn't trace back — the realised EPEX settlement was in fact removed after this check (and a correctness review) flagged it as a look-ahead violation, since it isn't known until after the auction the page exists to help decide.

8. Reproducibility

Public data only (ENTSO-E, Yahoo Finance). python main.py is the single command and runs the whole pipeline end-to-end from a committed few-MB parquet snapshot — **no API key required**: the LLM step loads from a committed cache (ai_logs/, complete enough to assess without rerunning) and the walk-forward backtests from committed result caches.

Deterministic — numpy/random/LightGBM seeds fixed, so results repeat byte-for-byte; dependencies are pinned in `requirements.txt`; `.env` is git-ignored and keys are read from environment variables, never committed. Measured ~27s on a cache-hit run, well inside the brief's ~10-minute budget. The data fetch scripts (`data/fetch_*.py`) are included for transparency but are not on the reproduction path.

9. Limitations & Next Step

The model is a deliberately transparent prototype, not a tuned production system: it carries no explicit outage/REMIT signal (excluded as too sparse pre-2021 to use honestly), and the NTC and CO2 inputs are approximations whose limits are documented rather than hidden. The natural next step reuses the existing grounded-LLM commentary engine: feed parsed outage and news signals into the same structured fact object, keeping the trader-facing narrative while adding the one input class the current feature set most lacks — without taking on the build risk of a live REMIT parser this week.

Full rationale, round-by-round build history, all figures, and the complete LLM/dashboard design discussion are in `full_report.pdf`. Reproduction: `python main.py` (single command, ~27s on a cache-hit run, no API keys required, deterministic).